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Studierichting: Informatica
Oefeningenreeks 4A nr. 40.1

$$\int \frac{dx}{\sqrt{x(4-x)}} = \int \frac{dx}{\sqrt{4x-x^2}} = \int \frac{dx}{\sqrt{-(x^2-4x)}}$$
$$= \int \frac{dx}{\sqrt{-(x^2-4x+4-4)}} = \int \frac{dx}{\sqrt{4-(x-2)^2}}$$

$t = x-2 \Rightarrow dt = dx$

$$\int \frac{dx}{\sqrt{4-(x-2)^2}} = \int \frac{dt}{\sqrt{4-t^2}} = \int \frac{dt}{\sqrt{4\left(1-\frac{t^2}{4}\right)}} = \int \frac{dt}{2\sqrt{1-\left(\frac{t}{2}\right)^2}}$$

$u = \frac{t}{2} \Rightarrow du = \frac{1}{2}dt \Rightarrow dt = 2du$

$$= \int \frac{2du}{2\sqrt{1-u^2}} = \int \frac{du}{\sqrt{1-u^2}} = \mathrm{Bgsin}(u) + c$$

Dus:

$$\int \frac{dx}{\sqrt{x(4-x)}} = \mathrm{Bgsin}\left(\frac{x-2}{2}\right) + c$$

References